Free vibration with viscous damping

The equation of motion can be written as

$$m\ddot{x} = -c\dot{x} - kx$$

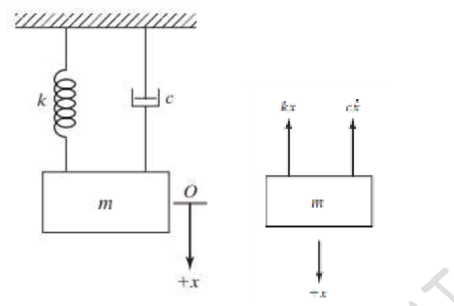
$$m\ddot{x} + c\dot{x} + kx = 0 \quad \ldots\ldots(1)$$

The solution of this equation can be assumed as

$$x(t) = Ae^{pt} \ldots\ldots(2)$$

$$m(Ap^2e^{pt}) + c(Ape^{pt}) + k(Ae^{pt}) = 0$$

$$Ae^{pt}(mp^2 + cp + k) = 0$$



From this, we can write the characteristic equation as $(mp^2 + cp + k) = 0$. The roots are

$$p_{1,2} = \frac{-c \pm \sqrt{c^2 - 4km}}{2m}$$

There are three possible cases depending upon the value of the discriminant.

Case (i): When $c^2 > 4km$ (The effect of damping force is much more predominant than that of inertia force and spring force)- OVER DAMPED. The roots are real and distinct

$$p_{1,2} = -\left(\frac{c}{2m}\right) \pm \sqrt{\left(\frac{c}{2m}\right)^2 - \left(\frac{k}{m}\right)}$$

Case (ii): When $c^2 = 4km$ - CRITICALLY DAMPED. The roots are real and equal

$$p_{1,2} = \frac{-c}{2m}$$

Case (iii): When $c^2 < 4km$ (The effect of damping force is much less than that of inertia force and spring force) -UNDER DAMPED. The roots are complex conjugates.

$$p_{1,2} = -\left(\frac{c}{2m}\right) \pm i\sqrt{\left(\frac{k}{m}\right) - \left(\frac{c}{2m}\right)^2}$$

Let us consider the critically damped system. The critical damping coefficient

$$c_c = \sqrt{4km} = 2m\sqrt{\frac{k}{m}} = 2m\omega_n$$

It is clear from the equation that the critical damping coefficient is a system property, which depends on the mass and stiffness of the system. Every system will have an associated critical damping coefficient. The actual damping in the system may or may not be equal to this value. The actual damping in the system can be expressed as a factor of the critical damping coefficent.

$$\frac{c}{c_c} = \zeta$$

Where ζ is a dimensionless number known as damping ratio. The damping coefficient of a system

$$c = \zeta\, c_c = 2m\zeta\omega_n \qquad \lll\ggg \qquad \frac{c}{2m} = \zeta\omega_n$$

For a critically damped sytem, $\zeta = 1$. The roots of the characteristic eqn are, $p_{1,2} = -\frac{c}{2m} = -\omega_n$.

The solution of the differential equation, $x(t) = (A_1 + A_2t)e^{-\omega_n t}$.

$$\dot{x}(t) = (A_1 + A_2t)(-\omega_n)e^{-\omega_n t} + A_2e^{-\omega_n t}$$

The arbitrary constants need to be evaluated from the initial conditions (when $t = 0$, $x = x_0$ and $\dot{x} = \dot{x}_0$)

$x_0 = A_1 \quad \lll\ggg \qquad \dot{x}_0 = A_1(-\omega_n) + A_2 \quad \lll\ggg \quad A_2 = \dot{x}_0 + x_0\omega_n$

$$x(t) = \{x_0 + (\dot{x}_0 + x_0\omega_n)t\}e^{-\omega_n t}.$$

For an overdamped system, $\zeta > 1$

$$p_{1,2} = -\left(\frac{c}{2m}\right) \pm \sqrt{\left(\frac{c}{2m}\right)^2 - \left(\frac{k}{m}\right)} = -\zeta\omega_n \pm \sqrt{(\zeta\omega_n)^2 - (\omega_n)^2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$$

The solution is

$$x(t) = A_1 e^{\left(-\zeta\omega_n + \omega_n\sqrt{\zeta^2-1}\right)t} + A_2 e^{\left(-\zeta\omega_n - \omega_n\sqrt{\zeta^2-1}\right)t} = e^{-\zeta\omega_n t}\left(A_1 e^{\omega_n\sqrt{\zeta^2-1}\,t} + A_2 e^{-\omega_n\sqrt{\zeta^2-1}\,t}\right)$$

$$\dot{x}(t) = A_1\left(-\zeta\omega_n + \omega_n\sqrt{\zeta^2-1}\right)e^{\left(-\zeta\omega_n + \omega_n\sqrt{\zeta^2-1}\right)t} + A_2\left(-\zeta\omega_n - \omega_n\sqrt{\zeta^2-1}\right)e^{\left(-\zeta\omega_n - \omega_n\sqrt{\zeta^2-1}\right)t}$$

The arbitrary constants need to be evaluated from the initial conditions (when $t = 0,\ x = x_0 \text{ and } \dot{x} = \dot{x}_0$)

$$x_0 = A_1 + A_2$$

$$\dot{x}_0 = A_1\left(-\zeta\omega_n + \omega_n\sqrt{\zeta^2-1}\right) + A_2\left(-\zeta\omega_n - \omega_n\sqrt{\zeta^2-1}\right) = -\zeta\omega_n(A_1 + A_2) + \omega_n\sqrt{\zeta^2-1}(A_1 - A_2)$$

$$\frac{\dot{x}_0 + \zeta\omega_n x_0}{\omega_n\sqrt{\zeta^2-1}} = A_1 - A_2$$

$$\frac{x_0}{2} + \frac{\dot{x}_0 + \zeta\omega_n x_0}{2\omega_n\sqrt{\zeta^2-1}} = A_1 \quad \lll\ggg \quad \frac{x_0}{2} - \frac{\dot{x}_0 + \zeta\omega_n x_0}{2\omega_n\sqrt{\zeta^2-1}} = A_2$$

For an underdamped system, $\zeta < 1$

$$p_{1,2} = -\left(\frac{c}{2m}\right) \pm i\sqrt{\left(\frac{k}{m}\right) - \left(\frac{c}{2m}\right)^2} = -\zeta\omega_n \pm i\,\omega_n\sqrt{1-\zeta^2} = -\zeta\omega_n \pm i\,\omega_d$$

Where $\omega_n\sqrt{1-\zeta^2} = \omega_d$, *damped frequecy*

The solution is

$$x(t) = A_1 e^{(-\zeta\omega_n + i\,\omega_d)t} + A_2 e^{(-\zeta\omega_n - i\,\omega_d)t} = e^{-\zeta\omega_n t}\left(A_1 e^{i\,\omega_d t} + A_2 e^{-i\,\omega_d t}\right)$$

$$= e^{-\zeta\omega_n t}\{A_1(\cos\omega_d t + i\sin\omega_d t) + A_2(\cos\omega_d t - i\sin\omega_d t)\}$$

$$= e^{-\zeta\omega_n t}\{(A_1 + A_2)\cos\omega_d t + (A_1 - A_2)i\sin\omega_d t\}$$

$$= e^{-\zeta\omega_n t}\{C_1\cos\omega_d t + C_2\sin\omega_d t\}$$

$$= e^{-\zeta\omega_n t}\{X_0\sin\varphi\cos\omega_d t + X_0\cos\varphi\sin\omega_d t\} \quad \ll\gg \quad X_0 = \sqrt{{C_1}^2 + {C_2}^2} \quad \ll\gg \quad \tan\varphi = \frac{C_1}{C_2}$$

$$= X_0\, e^{-\zeta\omega_n t}\sin(\omega_d t + \varphi)$$

Let us evaluate the arbitrary constants by substituting the initial conditions (when $t = 0,\ x = x_0 \text{ and } \dot{x} = \dot{x}_0$)

$$x(t) = e^{-\zeta\omega_n t}\{C_1\cos\omega_d t + C_2\sin\omega_d t\}$$

$$\dot{x}(t) = e^{-\zeta\omega_n t}\{-C_1\,\omega_d\sin\omega_d t + C_2\omega_d\cos\omega_d t\} - \zeta\omega_n e^{-\zeta\omega_n t}\{C_1\cos\omega_d t + C_2\sin\omega_d t\}$$

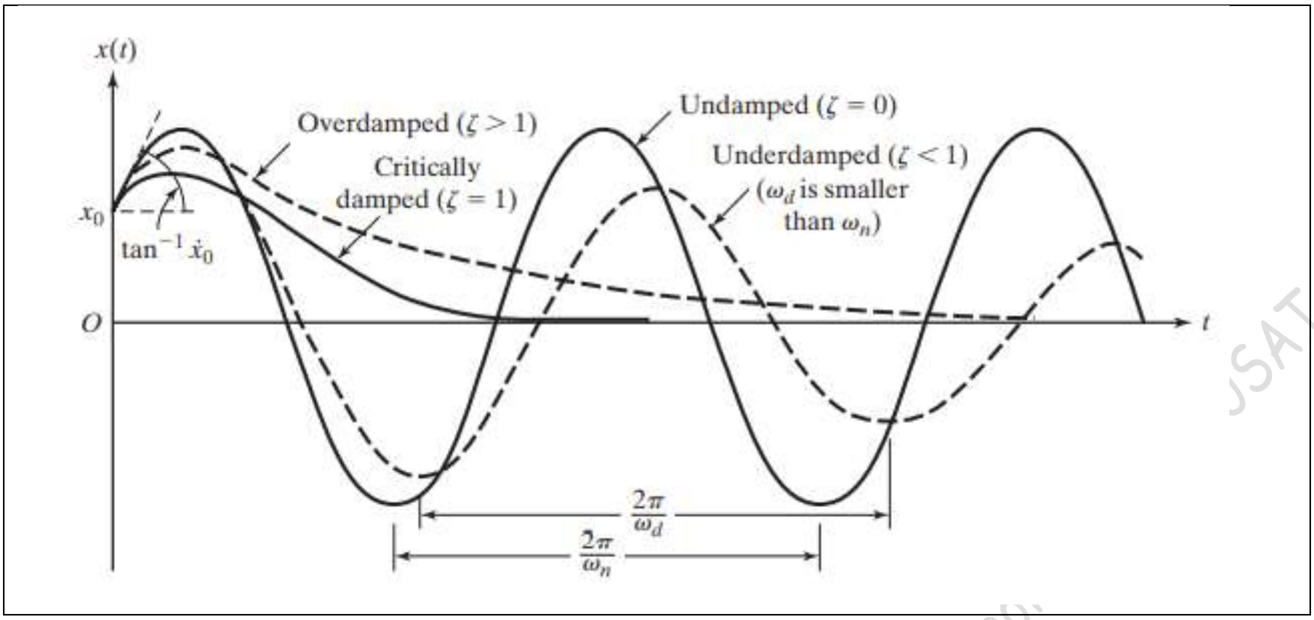
When $t = 0$,

$$x_0 = C_1$$

$$\dot{x}_0 = C_2\omega_d - \zeta\omega_n C_1 \quad \ll\gg \quad C_2 = \frac{\dot{x}_0 + \zeta\omega_n x_0}{\omega_d}$$

$$X_0 = \sqrt{{C_1}^2 + {C_2}^2} = \sqrt{{x_0}^2 + \left(\frac{\dot{x}_0 + \zeta\omega_n x_0}{\omega_d}\right)^2}$$

$$\tan\varphi = \frac{C_1}{C_2} = \frac{x_0}{\left(\frac{\dot{x}_0 + \zeta\omega_n x_0}{\omega_d}\right)}$$



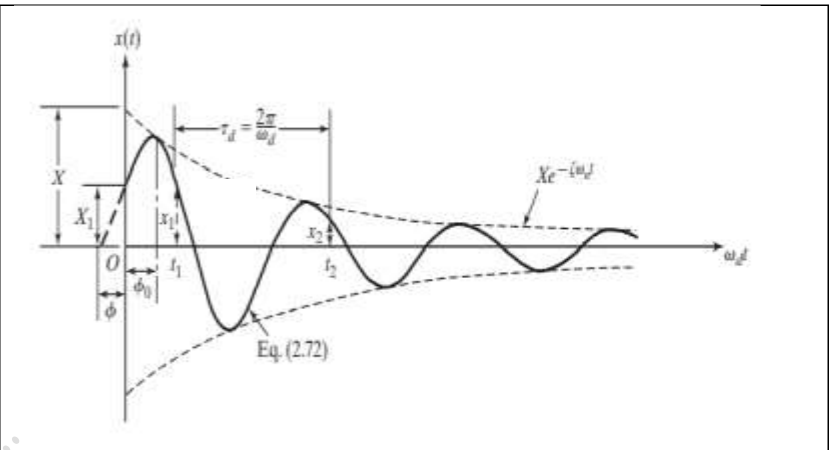
Logarithmic Decrement

Consider the response of an under-damped system. The displacement of the system at time t_1

$$x_1 = X_0 e^{-\zeta\omega_n t_1}\sin(\omega_d t_1 + \varphi)$$

The displacement of the system at time $t_2 = t_1 + \frac{2\pi}{\omega_d}$

$$x_2 = X_0 e^{-\zeta\omega_n t_2}\sin(\omega_d t_2 + \varphi) = X_0 e^{-\zeta\omega_n\left(t_1+\frac{2\pi}{\omega_d}\right)}\sin\left[\omega_d\left(t_1+\frac{2\pi}{\omega_d}\right)+\varphi\right]$$



$$\frac{x_1}{x_2} = \frac{X_0 e^{-\zeta\omega_n t_1}\sin(\omega_d t_1+\varphi)}{X_0 e^{-\zeta\omega_n\left(t_1+\frac{2\pi}{\omega_d}\right)}\sin\left[\omega_d\left(t_1+\frac{2\pi}{\omega_d}\right)+\varphi\right]} = \frac{e^{\zeta\omega_n\left(t_1+\frac{2\pi}{\omega_d}\right)-\zeta\omega_n t_1}\sin(\omega_d t_1+\varphi)}{\sin[2\pi+\omega_d t_1+\varphi]} = e^{\left(\frac{2\pi\zeta\omega_n}{\omega_d}\right)} = e^{\frac{2\pi\zeta\omega_n}{\omega_n\sqrt{1-\zeta^2}}} = e^{\frac{2\pi\zeta}{\sqrt{1-\zeta^2}}}$$

$$\ln\left(\frac{x_1}{x_2}\right) = \frac{2\pi\zeta}{\sqrt{1-\zeta^2}}$$

The natural logarithm of ratio of successive amplitudes is known as the logarithmic decrement (δ). The ratio of any two successive amplitudes will be a constant. If n cycles $x_1, x_2, \ldots, x_{n+1}$are taken,

$$\frac{x_1}{x_{n+1}} = \frac{x_1}{x_2}.\frac{x_2}{x_3}.\frac{x_3}{x_4}\ldots\ldots\ldots\frac{x_n}{x_{n+1}} = e^{\frac{2\pi\zeta}{\sqrt{1-\zeta^2}}}.e^{\frac{2\pi\zeta}{\sqrt{1-\zeta^2}}}.e^{\frac{2\pi\zeta}{\sqrt{1-\zeta^2}}}\ldots\ldots\ldots e^{\frac{2\pi\zeta}{\sqrt{1-\zeta^2}}} = e^{\frac{n2\pi\zeta}{\sqrt{1-\zeta^2}}}$$

$$\ln\left(\frac{x_1}{x_{n+1}}\right) = \frac{n2\pi\zeta}{\sqrt{1-\zeta^2}}$$

Energy dissipated in viscous damping

In a viscously damped system, the rate of energy dissipated $\left(\frac{dW}{dt}\right) = -damping\ force \times velocity$

$$= -c\dot{x} \times \dot{x} = -c\left(\frac{dx}{dt}\right)^2$$

Assuming a simple harmonic motion $x = X\sin\omega_d t \quad \lll\ggg \quad \dot{x} = X\omega_d \cos\omega_d t$

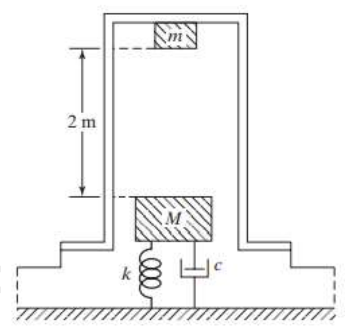
The energy dissipated in a cycle,

$$W = \int_0^{\frac{2\pi}{\omega_d}} dW = \int_0^{\frac{2\pi}{\omega_d}} -c\left(\frac{dx}{dt}\right)^2 dt = \int_0^{\frac{2\pi}{\omega_d}} -c(X\omega_d\cos\omega_d t)^2\, dt = -cX^2{\omega_d}^2 \int_0^{\frac{2\pi}{\omega_d}} \frac{1+\cos\ 2\omega_d t}{2}\, dt = -\pi c\omega_d X^2$$

Negative sign shows that energy is dissipated.

Example 1

The anvil of a forging hammer weighs $5000\ N$ and is mounted on a foundation that has stiffness of $5 \times 10^6 N/m$ and a viscous damping coefficient of $10000\ Ns/m$. During forging, the hammer weighing $1000\ N$, is made to fall from a height of $2m$ on to the anvil. If the anvil is at rest before impact, determine the response of the anvil after the impat. Assume the coefficient of restitution between the anvil and the hammer is 0.4



Since, the impact is semi-ealstic, the hammer will rebound after impact.

Mass of the anvil, $M = \frac{5000}{9.81} = 509.68\ kg$ The stiffness, $k = 5 \times 10^6 N/m$

The natural frequency of vibration of the anvil, $\omega_n = \sqrt{\frac{k}{M}} = \sqrt{\frac{5\times10^6 N/m}{509.68\ kg}} = 99.05\ rad/s$

The damping coefficient, $c = 10000\ Ns/m$

$$c = 2M\zeta\omega_n \quad \lll\ggg \quad \zeta = 0.099$$

Since, $\zeta < 1$, the system is under-damped.

The damped frequency, $\omega_d = \omega_n\sqrt{1-\zeta^2} = 98.56\frac{rad}{s}$

The response of the underdamped system is given by

$$x(t) = X_0\, e^{-\zeta\omega_n t}\sin(\omega_d t + \varphi)\ or\ \ e^{-\zeta\omega_n t}\{C_1\cos\omega_d t + C_2\sin\omega_d t\}$$

The arbitrary constants

$$C_1 = x_0 \quad \lll\ggg \quad C_2 = \frac{\dot{x}_0 + \zeta\omega_n x_0}{\omega_d}$$

Let us consider the vibration of the anvil after impact. The anvil starts its vibration from its equilibrium position $(x = 0)$ with a velocity which it acquires after the impact with the hammer.

The velocity of the hammer just before impact, $u = \sqrt{2gh} = \sqrt{2\times9.81\times2} = 6.26\ m/s$

The velocity of the hammer just before impact, $v = 0$

$$mu + Mv = mu' + Mv' \quad \lll\lll\ggg\ggg \quad \frac{1000}{9.81}\times6.26 = \frac{1000}{9.81}\times u' + \frac{5000}{9.81}\times v' \quad \lll\ggg \quad u' + 5v' = 6.26$$

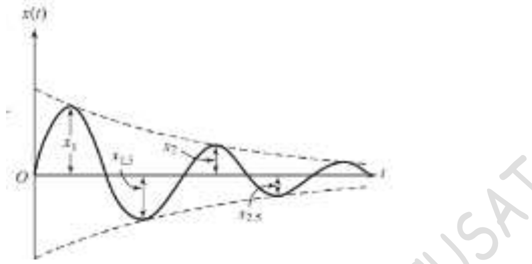
$$v' - u' = -e(v-u) \quad \lll\ggg \quad v' - u' = 2.504$$

Soving these two equations, we get, $v' = 1.46\ m/s$ and $u' = -1.04\ m/s$

Therefore the initial conditions are $x_0 = 0\ \ and\ \dot{x}_0 = 1.46\ m/s$

$$C_1 = 0 \quad \lll\ggg \quad C_2 = \frac{\dot{x}_0}{\omega_d} = \frac{1.46}{98.56} = 0.0148$$

$$x(t) = e^{-\zeta\omega_n t}\, C_2 \sin\omega_d t = 0.0148 e^{-9.806t} \sin 98.56t$$

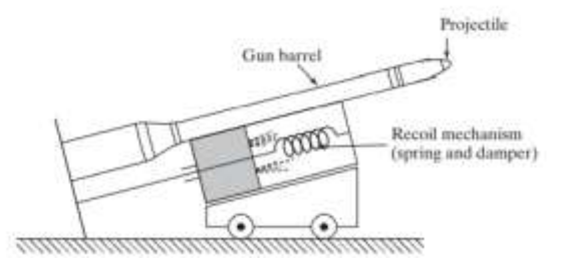
Example 2 An underdamped shock absorber is to be designed for a motorcycle of mass $200\,kg$. Find the necessary stiffness and damping constant of the shock absorber if the damped period of vibration is to be $2s$ and the amplitude is reduced to one-fourth in one half-cycle.	
Given, $m = 200\,kg$, $x_{1.5} = \frac{1}{4}x_1 \lll\ggg \frac{x_1}{x_{1.5}} = 4$ Since, the ratio of successive amplitudes are constant, $\frac{x_{1.5}}{x_2} = 4 \quad \lll\ggg \quad \frac{x_1}{x_2} = \frac{x_1}{x_{1.5}}.\frac{x_{1.5}}{x_2} = 16$	damped period $= T_d = 2s$ $\frac{2\pi}{\omega_d} = 2 \lll\ggg \omega_d = 3.14\,rad/s$ $\ln\left(\frac{x_1}{x_2}\right) = \ln 16 = 2.773$

$$\frac{2\pi\zeta}{\sqrt{1-\zeta^2}} = 2.773 \quad \lll\ggg \quad 4\pi^2\zeta^2 = 2.773^2(1-\zeta^2) \quad \lll\ggg \quad \zeta^2(39.478 + 7.689) = 7.689 \quad \lll\ggg \quad \zeta = 0.404$$

$$\omega_d = \omega_n\sqrt{1-\zeta^2} \quad \lll\lll\ggg\ggg \quad \omega_n = \frac{3.14}{\sqrt{1-0.404^2}} = 3.43\,rad/s$$

$$\omega_n = \sqrt{\frac{k}{m}} \quad \lll\lll\ggg\ggg \quad k = m\,\omega_n^{\;2} = 200 \times 3.43^2 = 2356.5\,N/m$$

Damping coefficient, $c = 2m\zeta\omega_n = 2 \times 200 \times 0.404 \times 3.43 = 554.288\,Ns/m$

Example 3 The schematic diagram of a large cannon is shown. In a particular case, the gun barrel and the recoil mechanism have a mass of $500\,kg$ with a recoil spring of stiffness $10000\,N/m$. The gun recoils $0.4\,m$ upon firing. Find (1) the critical damping coefficient of the damper (2) the initial recoil velocity of the gun.	
Mass of the system, $m = 500\,kg$ Stiffness of the spring, $k = 10000\,N/m$ Damping factor, $\zeta = 1$	Natural frequency of the system, $\omega_n = \sqrt{\frac{10000}{500}} = 4.472\,rad/s$

The critical damping coefficient, $C_c = 2\,m\,\omega_n = 2 \times 500 \times 4.472 = 4472\,Ns/m$

The displacement of the system, $x(t) = (A_1 + A_2 t)e^{-\omega_n t} = \{x_0 + (\dot{x}_0 + x_0\omega_n)t\}e^{-\omega_n t}$.

But, the recoil starts from its equilibrium position, $x_0 = 0$

$$\therefore x(t) = \dot{x}_0 t e^{-\omega_n t}$$

When the displacement is maximum, $\dot{x} = 0$

$$0 = \dot{x}_0 t(-\omega_n)e^{-\omega_n t} + e^{-\omega_n t}\dot{x}_0 \lll\ggg \omega_n t = 1 \quad \lll\ggg \ t = 0.224\,s$$

The maximum recoil displacement, $x_{max} = 0.4\,m = \dot{x}_0(0.224)e^{-4.4472\times0.224} \lll\ggg \dot{x}_0 = 4.83\,m/s$

#Exercise 1

The parmeters ofa single degree of freedom system are given by $m = 1\ kg, c = 5\frac{Ns}{m}, and\ k = 16\ N/m$. Find the response of the system for the following initial conditions

i) $x(0) = 0.1\ m\ and\ \dot{x}(0) = 2\ m/s$

ii) $x(0) = -0.1\ m\ and\ \dot{x}(0) = 2\ m/s$

#Exercise 2

The response of a single degree-of-freedom system that is initially displaced and released is given by $x(t) = 0.5e^{-6t}\sin(5t + 1.3333)m$. Determine the damping ratio, natural frequency, and the initial displacement of the system.

#Exercise 3

A torsional pendulum has a natural frequency of $200\ cycles/min$ when vibrating in vacuum. The mass moment of inertia of the disc is $0.2\ kg.m^2$. It is then immersed in oil and its natural frequency is found to be $180\ cycles/min$. Determine the damping constant. If the disc, when placed in oil, is given an initial dis[lacement of $2°$, find its displacement at the end of the first cycle.

#Exercise 4

Plot the root locus in scilab/matlab, of the system governed by the characteristic equation $3s^2 + cs + 27 = 0$, by varying the value of $c > 0$

#Exercise 5

Plot the root locus in scilab/matlab, of the system governed by the characteristic equation $s^2 + 16s + k = 0$, by varying the value of $k > 0$

#Exercise 6

Plot the root locus in scilab/matlab, of the system governed by the characteristic equation $ms^2 + 14s + 20 = 0$, by varying the value of $m > 0$

Reference:

Mechanical Vibrations, 6/e, Singiresu S. Rao, Pearson Global Edition